INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To,
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal
05th January 2026

Subject: Request for Approval of ISTE Projects

Respected Sir,

Greetings from the Indian Society for Technical Education, NITK Chapter.

The Indian Society for Technical Education (ISTE), NITK Chapter, undertakes a wide range of technical, research-oriented, and interdisciplinary projects through its eight Special Interest Groups (SIGs). These initiatives aim to enhance practical learning, promote innovation, and develop industry-relevant skills among students through hands-on implementation, design, and experimentation. Based on the consolidated component-wise requirements of all approved projects across the SIGs, the total estimated project expenditure amounts to ₹83,000. In addition, a contingency buffer of ₹2,000 has been included to accommodate unforeseen expenses such as minor price variations, replacement of defective components, or additional prototyping requirements. Accordingly, the total budget requested for the execution of all ISTE projects is ₹85,000.

We sincerely request approval for the above.

Regards,

Likith M.
(Convenor, ISTE NITK)

Dr. Vaishak Nair
(Faculty Advisor, ISTE NITK)

Prof. Govinda Raj Mandela
(Dean of Students Welfare, NITK)

Dr. Shashi Bhushan Arya
(PIC Technical Club, ISTE NITK)

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

SIG-wise Budget Allocation Summary

SIG (SUG)	Total Budget (₹)
Catalyst	₹3,200
Charge	₹20,600
Chronicle	₹7,000
Clutch	₹52,500
Create	₹0
Crypt	₹0
Credit	₹0
Concrete	₹0

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Catalyst Projects

Dear Sir,

With reference to the subject cited above, we seek permission to procure components required for the *Catalyst SIG* technical projects for the current academic year. Catalyst focuses on integrating chemical engineering fundamentals with sustainable technologies, electrochemical systems, computational modelling, and data-driven diagnostics. This year's project lineup includes the design and development of a Saltwater Lamp, real-time health monitoring of heat exchangers using PINNs and Hypernetworks, and waste heat recovery via an Organic Rankine Cycle (ORC). These projects collectively aim to promote innovation-driven learning and industry-relevant skill-building.

Permission is sought to purchase the necessary components to proceed with these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Saline Beacon

Mentors: Jonathan James, Adhil Ali, Ivine Binu
Mentees: John Jacob, Yallapa H.U, Aysha K.C

1. Problem

- Off-grid households and emergency situations often lack affordable, safe lighting. Traditional batteries are expensive and hazardous to dispose of, creating a need for sustainable power sources.

2. Objective

- Design a low-cost, portable saltwater-powered lamp.
- Achieve target lighting performance (~50 lumens, 6-hour runtime).
- Minimize electrode corrosion and ensure user safety with non-toxic components.

3. Methodology

- Electrode Optimization: Test Zn/Cu and Mg/C variations to maximize runtime and reduce corrosion.
- Circuit Design: Implement a buck/boost driver with <50 μA standby current for high efficiency.
- Enclosure Design: Create a compact, waterproof enclosure with replaceable cartridges.
- Testing: Evaluate voltage/capacity under series/parallel arrangements.

4. Expected Outcome

- A functional lamp delivering ~50 lumens for 6 hours using saltwater.
- A replaceable cartridge system enabling long-term sustainable use.
- Verified safety and durability for rural or emergency deployment.

5. Future Scope

- Scaling pilot production for deployment in disaster-prone areas.
- Developing modular cartridges for easier maintenance.
- Adapting the design into multi-cell power packs to charge small devices like phones.

6. Total Project Budget: Rs. 3,200

S No	Component	Cost
1	Container	Rs. 1300
2	Electrodes	Rs. 1000
3	Electrolytes	Rs. 200
4	LED	Rs. 200
5	Circuitry	Rs. 500
	Total	**Rs. 3200**

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

REAL-TIME HEALTH MONITORING USING PINN + HYPERNETWORK

Mentors: Rohit Singh, Yash Jain, Neha Ojha
Mentees: Abhishek Mahato, Diya Zacharia, Rudransh Pandey

1. Problem

- Energy-intensive chemical processes (e.g., Haber, Contact processes) release significant thermal energy (400−600◦C) into the environment, resulting in reduced efficiency and a higher carbon footprint.

2. Objective

- Model industrial processes to identify waste heat streams.
- Design and simulate an Organic Rankine Cycle (ORC) unit in DWSIM.
- Perform techno-economic analysis (Payback, NPV, IRR) to validate viability.

3. Methodology

- Baseline Simulation: Model hot sections of Haber/Contact processes in DWSIM.
- Fluid Screening: Test organic fluids (R245fa, Toluene, Isobutane) for efficiency.
- Parametric Study: Vary evaporating/condensing temperatures to optimize power output.
- Economic Analysis: Calculate LCOE and payback periods using Excel/Matlab.

4. Expected Outcome

- A validated simulation model achieving 7-15% cycle efficiency.
- Identification of optimal working fluids for specific industrial waste streams.
- A comprehensive Techno-Economic Evaluation demonstrating financial viability.

5. Future Scope

- Dynamic process modeling to evaluate ORC response under variable loads.
- Integration with renewable sources for hybrid power generation.
- Life Cycle Assessment (LCA) to quantify environmental impact.

6. Total Project Budget: Rs. 0 *(Software based project: DWSIM, Excel, Python)*

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

WASTE HEAT RECOVERY VIA ORGANIC RANKINE CYCLE (ORC)

Mentors: Sanjo Kedavath, Dathu Vikas, Shubhank Hiremath
Mentees: Soumili Dey, Antara Dongre, Chowhan Deepak

1. Problem

- Industrial heat exchangers suffer from fouling and efficiency loss. Traditional sensors cannot measure internal fields non-intrusively, making real-time health monitoring and predictive maintenance difficult.

2. Objective

- Develop a Physics-Informed Neural Network (PINN) as a digital twin of a heat exchanger.
- Use a Hypernetwork to map live boundary conditions to PINN weights.
- Enable real-time internal temperature estimation for fouling detection.

3. Methodology

- Dataset Generation: Train base PINNs on various boundary conditions.
- Hypernetwork Training: Optimize a network to predict PINN weights from sensor inputs.
- Deployment: Create an inference pipeline to estimate temperature fields from live data.
- Validation: Compare predictions against simulated plant data.

4. Expected Outcome

- Real-time prediction pipeline with <5% error.
- Early detection of fouling and efficiency drops.
- A dashboard enabling predictive maintenance to reduce downtime.

5. Future Scope

- Extend the model to boilers, condensers, and chemical reactors.
- Integrate with IoT dashboards for fleet-wide industrial deployment.

6. Total Project Budget: Rs. 0 *(Software based project: Python, DeepXDE, JAX)*

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

REAL-TIME AI & CFD HAZARD DETECTION SYSTEM

Mentors: Kshama Rai, Garima Yadav, Shanul Haque

Mentees: Pranjali Vishwakarma, Greeshma Prabhu, Nandini Eswaran

1. Problem

- Chemical plants are prone to critical hazards such as pressure drops, temperature spikes, and abnormal flow patterns. Traditional monitoring often reacts after an incident has escalated. There is a need for a proactive system that can not only detect these anomalies in real-time but also predict their propagation to ensure operator safety.

2. Objective

- Detect real-time hazards utilizing sensor data integration.
- Predict hazard evolution (e.g., gas dispersion, thermal propagation) using CFD-trained AI models.
- Trigger automated alarms and provide early warnings to operators.
- Visualize hazards in an interactive 3D dashboard for rapid decision-making.

3. Methodology

- CFD Simulation: Build plant geometries and run simulations to generate data on leak behavior and flow disturbances.
- Dataset Generation: Create structured datasets covering both normal and hazard operating conditions.
- Model Training: Train Machine Learning models to identify hazard signatures and estimate severity based on the CFD data.
- Dashboard Development: Build a real-time Streamlit dashboard for alarms and 3D visualization.
- Integration: Deploy the full pipeline to test detection and prediction capabilities.

4. Expected Outcome

- A fully functional AI-driven hazard detection system capable of identifying anomalies early.

- A 3D interactive dashboard that visualizes hazard propagation for operators.
- A working demo integrating Computational Fluid Dynamics (CFD) with real-time AI inference.

5. Future Scope

- Generalize the framework to high-hazard sectors like refineries and gas storage facilities.
- Extend CFD coverage to more complex geometries and chemical handling scenarios.
- Integrate real-plant sensor data for live industrial deployment.

6. Total Project Budget: Rs. 0 (Software based project: ANSYS, Python, Streamlit)

Catalyst Budget Distribution

Catalyst - Saline Beacon

S No	Component	Cost
1	Container	Rs. 1300
2	Electrodes	Rs. 1000
3	Electrolytes	Rs. 200
4	LED	Rs. 200
5	Circuitry	Rs. 500
	Total	**Rs. 3,200**

Total budget allocated for the SIG Catalyst - Rs 3,200

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Charge Projects

Dear Sir,

With reference to the subject cited above, we seek permission to procure components required for the CHARGE SIG technical projects for the current academic year. The SIG focuses on advancing student competence in embedded systems, digital hardware design, robotics, and real-time signal processing through hands-on engineering projects. This year's project lineup includes the development of an Autonomous Navigation Robot using LiDAR-based SLAM , a Gesture-Controlled Bluetooth Speaker employing on-device machine learning , a RISC-V Multicore Processor prototyped on FPGA , an Offline FPGA-based Speech-to-Speech Translator integrating DSP and embedded ML models , and an AES-256 Encryption System implemented from first principles. These projects collectively aim to expose students to industry-relevant technologies, enhance practical design skills, and promote innovation-driven learning at NITK.
Permission is sought to purchase the necessary components to proceed with these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: AUTONOMOUS NAVIGATION ROBOT

Mentors: Kaif , Nisha , Venkatesh
Mentees: Chinmay , Sanjeetha , Aditya Khankar , Shreyash

1. Problem

Autonomous indoor navigation is a key challenge in robotics. Robots require accurate maps and localization to operate in unknown environments. Students must learn SLAM, sensor fusion, and path planning using practical implementation.

2. Objective

- Implement a robot capable of **Simultaneous Localization and Mapping (SLAM)**.
- Fuse data from LiDAR, wheel encoders, and IMU using an **Extended Kalman Filter**.
- Perform autonomous path planning and obstacle avoidance.
- Build a functional prototype using ROS and onboard computing.

3. Methodology

- Use ROS with 2D LiDAR GMapping for SLAM.
- Implement EKF for localization.
- Use global and local planners for navigation.
- Build chassis, motor drivers, and Raspberry Pi compute unit.
- Test navigation in simulated and real environments.

4. Expected Outcome

- Robot capable of mapping unknown indoor spaces.
- Accurate localization using fused sensors.
- Obstacle avoidance with autonomous navigation.

5. Future Scope

- Integration of 3D sensors for full-space mapping and improved dynamic replanning algorithms..
- Multi-robot mapping and coordination.
- Deployment in delivery/warehouse applications.

6. Total Project Budget: ₹16,000

Components	Price
YDLiDAR G2 360 degree 2D LiDAR	Rs. 10,000
Raspberry Pi4	Rs. 6,000

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: GESTURE-CONTROLLED BLUETOOTH SPEAKER

Mentors: Vinay , Mahadev , Shreesha , Siri

Mentees: Aditya Khankar , Devendranath Reddy , Partha Sarathi , Vrushank

1. Problem

Traditional audio interfaces require physical interaction, which is inconvenient in kitchens, workshops, medical labs, or accessibility scenarios. A touch-free user interface can improve hygiene, comfort, and accessibility.

2. Objective

- Build a portable Bluetooth speaker controlled entirely via hand gestures.
- Use ESP32-CAM for vision-based gesture recognition.
- Deploy lightweight ML models using TensorFlow Lite / Edge Impulse.
- Implement Bluetooth AVRCP commands for playback control.

3. Methodology

- Collect gesture dataset and label training samples.
- Train CNN using TensorFlow Lite / Edge Impulse.
- Implement on-device inference on ESP32-CAM.
- Integrate Bluetooth AVRCP for playback control.
- Assemble hardware enclosure and speaker module.

4. Expected Outcome

- Working prototype capable of gesture-based play/pause, track change, and volume control.
- On-device ML gesture recognition with >80% accuracy.

5. Future Scope

- Add voice commands and multimodal control.
- Extend to smart-home automation (fans, lights).
- Improve model robustness with larger datasets.

6. Total Project Budget: ₹3100-3500

Components	Price
ESP32 - CAM Module (Qty:2)	Rs. 1600
Bluetooth Speaker/Module	Rs. 1500

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: RISC-V MULTICORE PROCESSOR

Mentors: Vaishnav K , Sowmith

Mentees: Rihan , Sai Abhinay , Sanjeetha , Rhea Mishra , Neria

1. Problem

Modern applications such as graphics and AI require parallel computation. Students lack exposure to multicore CPU design, pipelining, FPU integration, and FPGA prototyping. This project provides hands-on experience in building a multicore RISC-V processor.

2. Objective

- Design and implement a multicore RISC-V processor in Verilog.
- Integrate a Floating-Point Unit (F-extension).
- Implement SIMD-style matrix operations.
- Build complete pipelined datapath with hazard handling.
- Prototype on FPGA with UART-based PC communication.

3. Methodology

- Study RISC-V ISA, GPU/SIMD concepts, and F-extension.
- Design datapath, pipeline stages, FPU, and register file.
- Implement hazard detection + forwarding.
- Add multicore dispatcher and shared-memory protocol.
- Simulate using ModelSim/Icarus and test on FPGA.

4. Expected Outcome

- Functional multicore RISC-V processor.
- Verified FPU and SIMD matrix operations.
- FPGA prototype with UART communication.

5. Future Scope

- Add cache hierarchy for faster memory access.
- Develop compiler support for high-level programming.
- Benchmark against commercial GPUs/CPUs.

6. Total Project Budget: ₹200 - 300

Components	Price
CP2102 USB - UART Converter	Rs. 200 - 300

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: OFFLINE SPEECH-TO-SPEECH TRANSLATOR ON FPGA

Mentors: Kshitij , Hafiz , Manan
Mentees: Rafiq , Swetha , Sharan

1. Problem

Existing translation tools depend heavily on cloud services. Offline translation is essential for low-connectivity environments, embedded systems education, and assistive communication devices.

2. Objective

- Build a fully offline speech translator using FPGA + MCU/SoC.
- FPGA performs DSP pipeline: VAD → FFT → Mel Filterbank.
- MCU performs ASR → MT → TTS using lightweight ML models.
- Achieve near real-time translation of short phrases.

3. Methodology

- Study DSP fundamentals (FFT, Mel) and embedded ASR/TTS models.
- Implement FPGA audio input + DSP pipeline.
- Deploy lightweight ML models on Raspberry Pi or MCU.
- Design FPGA↔MCU communication protocol (SPI/I²C).
- Integrate full pipeline from input speech → translated output .

4. Expected Outcome

- A functioning offline translator capable of converting speech from one language to another.
- Demonstration of FPGA DSP + embedded ML collaboration.
- Low-latency translation of predefined phrases.

5. Future Scope

- Add more languages and continuous speech support.
- Improve TTS naturalness and ASR accuracy and Integrate LCD/mobile UI for better usability.

6. Total Project Budget: ₹800–1200

Components	Price
Audio Codec(WM8731)	Rs.500 - 700
Microphone or Speaker	Rs. 300 - 500

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: AES-256 ENCRYPTION SYSTEM

Mentors: Vishal , Nitish , Rishi

Mentees: Pratheek , Rohini , Aditya Mondel , Shreyash

1. Problem

Modern communication systems require secure encryption methods. Students often treat cryptography as abstract theory without hands-on understanding. There is a need for a practical project that teaches modern symmetric-key encryption by implementing AES-256 from scratch.

2. Objective

- Implement the AES-256 block cipher entirely in Python.
- Build core AES transformations: SubBytes, ShiftRows, MixColumns, AddRoundKey.
- Implement 256-bit key expansion (14 rounds).
- Validate correctness against industry-standard libraries.
- Create a functional tool to encrypt/decrypt text or files.

3. Methodology

- Study NIST AES-256 specification.
- Implement SubBytes, ShiftRows, MixColumns, AddRoundKey.
- Develop the 14-round key schedule.
- Build encryption/decryption modules.
- Verify correctness using PyCryptodome/OpenSSL.
- Develop a CLI tool for file/text encryption.

4. Expected Outcome

- A functional AES-256 encryption system.
- Educational demonstration of internal AES workflow.
- Command-line encryption tool suitable for teaching and demos.

5. Future Scope

- Add AES decryption and authenticated modes (CBC, CTR, GCM).
- Optimize code using NumPy or C/C++ acceleration.
- Explore FPGA/Hardware implementation.
- Integrate into GUI-based secure storage/chat applications.

6. Total Project Budget: ₹0 (Software used: Python — open-source libraries)

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Chronicle Projects

Dear Sir,

With reference to the subject cited above, we seek approval to proceed with the semester initiatives for Chronicle, the literary and art SIG of ISTE. It focuses on narrative experimentation across various mediums. For the current academic session, we have planned a diverse variety of projects, including an Interactive Visual Novel that explores the nuances of student life and an AI-Powered Visual Comic designed to showcase club unity through generative technology. Additionally, we are organizing Feud Escape, a competitive event blending mental and physical challenges, alongside a Narrative Short Film project aimed at honing technical production and storytelling skills. These initiatives are designed to bridge the gap between traditional literary arts and innovative digital media, fostering creativity within the student community.

Permission is sought to proceed with the development and implementation of these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Chronicle Project: A Day in the Life

Mentors
Nisha
Arjun
Ivine
Siddarthan

Mentees
Dhilleswari
Jeevani
Deekshitha

About the Project
This project is an interactive, web-based visual novel that invites players to navigate the daily life of an NITK student through a dynamic, choose-your-own-adventure format. By blending traditional narrative techniques with new-age digital media, the game encourages players to explore diverse storytelling modes while making critical decisions that lead to multiple unique endings. The experience is designed to address complex themes like academic rigor, peer pressure, and self-discovery through a light-hearted and humorous lens. By transforming serious college challenges into an engaging and relatable game, the project aims to spark open conversations and promote "out-of-the-box" thinking. Ultimately, it serves as an intriguing entry point into the world of interactive media, fostering empathy and reflection on the modern student experience.

Components Required: None

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Chronicle Project: Feud Escape

Mentors
Aaryan
Sahitya
Johan
Priyanshu

Mentees
Sanjeev
Kalyani
Soumili

About the Project
Feud Escape is a high-energy, competitive event designed to bridge the gap between mental agility and tactical execution. This innovative project challenges teams to transition from a Family Feud stage environment—where they must master rapid-fire survey questions—to a physical Escape Room sequence requiring intense logic and collaboration. By rewarding survey performance with time advantages in the puzzle round, the event creates a high-stakes atmosphere that tests decision-making under pressure. Juniors gain hands-on event management experience by designing themes, constructing complex clue structures, and managing live game flow. Ultimately, the project serves as a dynamic platform for enhancing student engagement, fostering critical reasoning, and building leadership skills through friendly, fast-paced competition.

The following are the estimation details required for completing the ISTE Chronicle Project: Feud Escape

S. No	Component	Amount
1	Printing	Rs. 1000
2	Stickers	Rs. 1000
3	Colour Markers	Rs. 1000
	TOTAL	Rs. 3000

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
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National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Chronicle Project: ISTEVERSE Sequel

Mentors
Hima
Yash
Ronak
Bhavya

Mentees
Sanjeev
Kalyani
Soumili

About the Project
ISTEverse is a creative storytelling project designed to serve as a high-impact promotional tool for ISTE, showcasing the club's diverse ecosystem through a visual comic format. The project features eight central characters acting as mascots for each of the eight Special Interest Groups (SIGs), embodying their unique technical strengths and distinct club identities. By leveraging AI-powered tools such as Leonardo AI and ChatGPT alongside Canva, the team constructs an immersive narrative set in the mythical "ISTE land" to symbolize club unity. The storyline follows eight ancient forces seeking a hero to protect their realm, effectively mirroring the recruitment process and growth of new club recruits. This initiative culminates in both digital slideshows and physical copies, serving as a multi-media platform for club outreach and member engagement. Juniors gain hands-on experience in prompt engineering, visual composition, and narrative design while mastering professional comic creation software.

The following are the estimation details required for completing the ISTE Chronicle Project: ISTEVERSE Sequel

S. No	Component	Amount
1	Printing	Rs. 2000
	TOTAL	Rs. 2000

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

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17th November 2025

ISTE Chronicle Project:Short Film

Mentors	**Mentees**
Hafiz	Alphin
Chinmay	Hitee
Kaif	Kruthika

About the Project

This short film project is dedicated to creating an engaging 5-10 minute narrative while developing comprehensive, hands-on skills in filming, lighting, sound, and post-production. The primary project goals involve executing a full production workflow, from brainstorming and storyboard creation to the integration of innovative interactive video elements. Through collaborative learning and assigned team roles, the project aims to foster technical excellence and creative depth in storytelling. The future scope is designed to build a professional foundation for festival-level multimedia submissions, ensuring the final cut is pitch-ready for exhibitions. Key learning outcomes include mastering quality review processes and gaining proficiency in advanced visual effects and sound design to showcase technical innovation

The following are the estimation details required for completing the ISTE Chronicle Project: Short Film

S. No	Component	Amount
1	Costume alterations	Rs. 1000
2	Poster & Miscellaneous	Rs. 1000
	TOTAL	Rs. 2000

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Clutch Projects

Dear Sir,

With reference to the subject cited above, we wish to purchase components for the projects under Clutch, the Mechanical SIG of ISTE. It focuses on diverse mechanical domains, ranging from automobiles and aerodynamics to robotics and dynamics. This semester, we are undertaking projects based on both experimentation and simulation, including the development of a Fire Fighting Robot, an autonomous Delivery Drone, an Indoor Airship, and the Neoleg Exoskeleton. These initiatives aim to foster technical expertise by building working models and validating designs through rigorous analysis.

Permission is sought to purchase the necessary components to proceed with these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

8th December 2025

ISTE Clutch Project: Aircraft Bearing Bracket

Mentors	**Mentees**
Gulzaar	Ishita
Jeeva	Kruthika
Kailash	Shubham

About the Project
A simulation-based project focused on digitally redesigning an aircraft bearing bracket using topology optimization and validating its structural health through vibration (modal) analysis.

Objective
- Reduce bracket weight while ensuring strength and reliability.
- Use vibration analysis to assess structural health and detect potential fatigue.

Methodology
- Create CAD model and define optimization space.
- Perform static structural FEA and topology optimization in ANSYS.
- Run modal analysis to compare natural frequencies before and after optimization.
- Evaluate aerospace materials (Ti-6Al-4V, Maraging Steel, 316L).
- Validate results for stress, vibrational performance, and safety factor

Components / Tools Required
- ANSYS 2025
- Fusion 360 / SolidWorks

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INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
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Surathkal

8th December 2025

ISTE Clutch Project: Fire Fighting Robot

Mentors
Tejaswini
Nitesh
Johan
Siddharth

Mentees
Abhinandan
Aditya

About the Project

A compact autonomous robot capable of detecting fire, navigating obstacles, and extinguishing flames using a pump-nozzle mechanism.

Objective

- Detect fire using flame/temperature sensors.
- Navigate autonomously with obstacle avoidance.
- Spray water/foam through a robotic arm mechanism.
- Provide optional manual override.

Methodology

- Arduino UNO controls motors, sensors, and pump.
- Flame/IR sensor detects fire and initiates motion.
- Ultrasonic sensor handles obstacle avoidance.
- Servo-based nozzle aims and sprays water.

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INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

The following are the estimation details required for completing the ISTE Clutch Project:
Fire fighting robot

S. No	Component	Amount
1	Chassis	Rs. 2300
2	DC drive motors + gearheads	Rs. 2000
3	Wheels	Rs. 1200
4	Pump (mini water pump)	Rs. 1500
5	Nozzle + servo for aiming	Rs. 900
6	Battery (Li-Po)	Rs. 2000
7	Misc (bearings, fasteners, mounts)	Rs. 1000
	TOTAL	Rs. 11,500

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

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8th December 2025

ISTE Clutch Project: Autonomous Delivery Drone

Mentors	**Mentees**
Ashmit	Yaagh
Srujan	Azeem
Wrick	Sreenandh

About the Project

An autonomous quadcopter designed for GPS-based delivery of lightweight parcels (documents, medicines) across short distances—optimized for NITK Campus.

Objective

- Build a quadcopter capable of autonomous navigation.
- Implement GPS-based path following and obstacle avoidance.
- Develop a servo-actuated payload drop mechanism.

Methodology

- Design airframe and analyze payload capacity using ANSYS.
- Integrate Pixhawk, Raspberry Pi 5, GPS, motors, and servo release system.
- Use LiDAR/Rangefinders for obstacle detection.
- Write Python scripts for take-off, navigation, altitude hold, and parcel drop.
- Configure Mission Planner for tuning and flight parameters.
- Perform flight testing and refine design.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

The following are the estimation details required for completing the ISTE Clutch Project: Autonomous Delivery Drone

S. No	Component	Amount
1	Drone frame (carbon/CF-reinforced)	Rs. 3000
2	Propellers (set)	Rs.1000
3	Landing gear	Rs.2000
4	Payload release servo & mechanism	Rs.1000
5	Battery mounting & vibration dampers	Rs.5000
6	Misc (fasteners, 3D-printed mounts)	Rs.2500
	TOTAL	Rs. 14,500

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
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8th December 2025

ISTE Clutch Project:Airship

Mentors	**Mentees**
Rajitha	Yashika
Aditya	Gamana
Veeresh	Om

About the Project
A lightweight, helium-filled indoor airship designed for silent aerial surveillance with live video streaming using an ESP32-CAM.

Objective
- Build a remotely controlled LTA (Lighter-Than-Air) platform.
- Enable real-time video streaming for surveillance.
- Ensure safe, silent, long-endurance indoor operation.

Methodology
- Perform buoyancy and weight budget calculations.
- Model envelope, frame, and gondola in CAD.
- Fabricate Mylar envelope and carbon-fiber frame.
- Install motors, servos, ESP32-CAM, and RC receiver.
- Program video streaming server and test controls.
- Conduct flight tests after helium inflation.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

The following are the estimation details required for completing the ISTE Clutch Project: Airship

S. No	Component	Amount
1	Mylar envelope (material + sealing)	Rs. 2000
2	Carbon-fiber rods (frame)	Rs.1000
3	Gondola (3D printed)	Rs.2000
4	Propulsion mounts & bearings	Rs.1000
5	contingency and electronics	Rs.3000
6	Misc (fasteners, 3D-printed mounts)	Rs.2500
	TOTAL	Rs. 11,500

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

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Surathkal

8th December 2025

ISTE Clutch Project: Motogp

Mentors	**Mentees**
Manoj	Shubham
Ruthvik	Anagha
Dinesh	Raghava
Vivek	

About the Project

A simulation-driven project focused on designing a high-performance MotoGP racing motorcycle. This project does not involve physical fabrication; instead, it emphasizes CAD modeling, aerodynamics, structural analysis, and performance simulation using advanced CAE tools.

Objective

- To design and develop a MotoGP-style racing bike that adheres to the competition guidelines of **MotoGP FMSCI Student Bike Racing**.
- To validate performance, safety, and structural reliability through simulations.

Methodology

- Create a detailed 3D CAD model of the motorcycle including chassis, fairing, suspension geometry, and engine packaging.
- Perform aerodynamic simulations (CFD) to optimize drag, lift, and stability.
- Conduct structural analysis on chassis, swingarm, and key load-bearing components.
- Carry out weight distribution, handling geometry, and performance studies.
- Validate the design against MotoGP student competition rules and refine iteratively.
- Document results and prepare final concept presentation for competition submission.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

8th December 2025

ISTE Clutch Project: Neoleg

Mentors	**Mentees**
Vinay	Shravani
Bhoomika	Siddharth
Rishita	Param
Shreyas	Shamik
Shrujan	

About the Project

A design-focused mechanical engineering project aimed at developing a passive lower-limb exoskeleton that assists human movement without motors or electronics. The project emphasizes CAD modeling, human–exoskeleton alignment mechanisms, load-transfer optimization, and simulation-based validation rather than fabrication.

Objective

- To design a passive exoskeleton that reduces physical load on the wearer during tasks such as walking, lifting, or industrial work.
- To study human joint biomechanics and create a mechanically compatible system that supports safe and efficient movement.

Methodology

- Study human lower-limb anatomy and joint ranges to identify mechanical alignment requirements.
- Design a multi-joint passive exoskeleton in CAD with hip, knee, and ankle alignment mechanisms.
- Perform static and dynamic simulations to evaluate load transfer, comfort, and joint compatibility.
- Prepare a fully validated concept model with mechanical drawings and simulation reports.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

The following are the estimation details required for completing the ISTE Clutch Project: Neoleg

S. No	Component	Amount
1	Aluminium square/round rods (structural frame)	Rs. 2000
2	Carbon-fiber tubes / lightweight alloy rods	Rs.1000
3	3D-printed parts (hip/knee joints, sliders, brackets)	Rs.2000
4	Bearings / bushings (for pivot joints)	Rs.1000
5	Fasteners (M4–M8 bolts, nuts, washers)	Rs.3000
6	Straps & buckles (industrial-grade velcro and nylon straps)	Rs.2500
7	Foam padding / ergonomics	Rs.500
8	Springs / elastic bands	Rs.2500
9	Misc (adhesives, sandpaper, paint, cable ties, prototype consumables)	Rs.500
	TOTAL	Rs. 15,000

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Concrete Projects

Dear Sir,

With reference to the subject cited above, we seek permission to proceed with the projects under Concrete, the Civil Engineering SIG of ISTE. It focuses on structural integrity, geotechnics, and smart infrastructure. This semester, we are undertaking projects based on both experimentation and advanced data simulation. The project lineup includes the Pavement Condition Index Prediction using Machine Learning to reduce manual survey costs, and an Earthquake Waveform Prediction system that assesses building safety (Safe/Unsafe) using IS 1893 and IS 456 codes. We are also conducting a Seismic Hazard Zonation study to identify fault-sensitive zones in the Himalayas, alongside a structural analysis of Isolated RC Footings on improved soil to demonstrate the benefits of stabilization. These initiatives aim to foster technical expertise by building predictive models and validating structural designs through rigorous analysis

Permission is sought to proceed with the development and implementation of these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: PAVEMENT CONDITION INDEX PREDICTION USING MACHINE LEARNING

Mentors: Tushar, Piyush, Vijay

1. Problem Traditional Pavement Condition Index (PCI) determination relies on resource-intensive manual surveys, which often makes assessments subjective and costly. There is a critical need to address these challenges to reduce both time and cost while increasing assessment accuracy and objectivity in road maintenance.

2. Objective

- Collect and preprocess pavement condition data, including traffic, climate, and various distress types.
- Train and validate machine learning models to predict Pavement Condition Index (PCI) values with high accuracy.
- Automate the classification of pavement segments into categories like Excellent, Good, Fair, and Poor using predictive analytics.
- Integrate results into a GIS platform to support visual, data-driven decision-making in road maintenance.

3. Methodology

- **Data Collection & Engineering:** Collect historical PCI values, distress metrics (cracks, rut depth, potholes), traffic volumes, and climatic factors. Perform feature engineering to derive attributes such as pavement age and climate severity index.
- **Model Development:** Implement machine learning algorithms including Random Forest, Gradient Boosting, XGBoost, and LSTM for regression to predict PCI values. Use SVM, Decision Tree, and Random Forest for classification tasks.
- **Evaluation:** Assess performance using R, RMSE, and MAE for regression, and accuracy, precision, recall, and F1-score for classification.

4. Expected Outcome

- A functional user interface via Streamlit for data input and visualization, empowering non-technical stakeholders.
- Automated PCI calculation enabling proactive maintenance and resource optimization for road authorities.

5. Future Scope

- Integration with real-time IoT sensor data for dynamic monitoring.
- Incorporation of satellite imagery and deep learning for advanced analytics.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: EARTHQUAKE WAVEFORM PREDICTION & BUILDING SAFETY ASSESSMENT USING ML

Mentors: Sushanth, Amarnath, Charan, Darshan

1. Problem Traditional earthquake monitoring systems focus primarily on detection and shaking intensity, lacking prescriptive advice for individuals and authorities. Additionally, in developing countries, access to advanced structural software like ETABS is limited, creating a need for low-cost, accessible safety evaluation tools.

2. Objective

- Develop a real-time, low-cost earthquake early-warning system that predicts earthquake intensity and evaluates structural safety.
- Predict key seismic parameters such as Peak Ground Acceleration (PGA), Peak Ground Velocity (PGV), and generalized waveforms of P and S-waves.
- Assess structural impact using IS 1893 and IS 456 codes to determine if a building is Safe, Partially Safe, or Unsafe.
- Provide a clear verdict to guide immediate measures like evacuation or reinforcement.

3. Methodology

- **Data Processing:** Collect seismic data from IRIS/FDSN in miniSEED format and use ObsPy to convert it to tabular form, separating P-wave and S-wave components.
- **Intensity Prediction (Model 1):** Train regression models (XGBoost, LSTMs, etc.) on processed features to predict PGA, PGV, and intensity.
- **Waveform Generation (Model 2):** Use sequence-to-sequence models (LSTMs, GRUs, or CVAE) to predict generalized S-wave envelopes based on incoming P-wave data.
- **Structural Assessment:** Calculate base shear using IS 1893 formulas and compare it against column shear capacity derived from IS 456 provisions.

4. Expected Outcome

- A deployed machine learning system that predicts seismic parameters and generates realistic waveform approximations.
- A simple, low-cost interactive web application that provides actionable safety verdicts (Safe, Partially Safe, Unsafe).
- Reduced casualties and property loss through immediate, predictive, and prescriptive safety advice.

5. Future Scope

- Extend the dataset to thousands of earthquakes worldwide for higher accuracy.
- Integrate with IoT-based vibration sensors for continuous building monitoring.
- Incorporate extended structural metrics like displacement and inter-story drift.
- Implement reinforcement learning for adaptive model improvement.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: ISOLATED RC FOOTING ON NORMAL & IMPROVED SOIL

Mentors: Ayush, Aarshi, Gajendra, Sanjana

1. Problem Weak soil causes excessive settlement and possesses low bearing capacity, which makes construction costly if deep foundations are required to mitigate these issues.

2. Objective

- Evaluate the improvement in soil properties after adding required particle-sized minerals based on particle packing analysis.
- Compute allowable bearing capacity and settlement for both normal and improved soils.
- Design an isolated reinforced concrete footing for a given column load on both soil types to compare performance.

3. Methodology

- **Testing & Analysis:** Conduct soil property testing (unit weight, moisture, specific gravity) and compaction/strength testing (Standard Proctor, Triaxial).
- **Particle Packing:** Perform Particle Size Distribution Analysis using Sieve Analysis and EMMA Software to identify the required mineral for optimal packing.
- **Stabilization:** Improve soil by adding the required minerals and curing (if cementitious) for 28 days.
- **Design & Simulation:** Calculate bearing capacity using IS Code 6403 and perform structural design of the footing using the Limit State Method, STAAD.Pro, and Excel.

4. Expected Outcome

- Demonstration of how soil stabilization can improve bearing capacity and reduce settlement.
- A sustainable, low-cost alternative to deep foundations that saves construction materials.
- A complete design report comparing normal vs. improved soil performance with reinforcement detailing.

5. Future Scope

- Testing other admixtures such as lime and nano-silica.
- Expanding the research to include raft or pile foundations.
- Conducting life-cycle cost analysis for large-scale field implementation.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: SEISMIC HAZARD ZONATION AND CLUSTER ANALYSIS ACROSS THE NW HIMALAYA

Mentors: Sahitya, Shreyas & Vinit

1. Problem: Current seismic maps provide broad zones that fail to capture local risk variations in the Northwest Himalaya, a region highly prone to earthquakes due to complex tectonic features like the Main Central Thrust and Main Boundary Thrust38.

2. Objective

- Refine seismic hazard zonation by using data-driven clustering of earthquakes based on magnitude, depth, and fault proximity.
- Identify seismic hotspots and fault-sensitive zones to provide better insights for infrastructure safety and disaster management.
- Analyze the influence of fault proximity and depth on seismic activity.

3. Methodology

- **Data Preparation:** Collect and clean earthquake catalogue data and digitize major faults (MCT, MBT, HFT) in QGIS.
- **Feature Engineering:** Compute shortest distances from events to faults and normalize features (latitude, longitude, depth, magnitude).
- **Clustering:** Apply the Elbow Method to select the optimal number of clusters and use K-Means to group epicenters.
- **Hazard Analysis:** Compute Gutenberg-Richter a and b values for each cluster to interpret recurrence patterns and likelihood of large events.

4. Expected Outcome

- A refined hazard zonation map that highlights seismic hotspots and local variations.
- Identification of clusters with low b-values and high fault sensitivity.
- Recommendations for planning and risk mitigation based on validated results against IS 1893:2016.

5. Future Scope

- Integration of real-time seismic data and advanced machine learning models.
- Incorporation of geodetic data (GPS/InSAR) to improve hazard accuracy.
- Extension of the framework to other Himalayan regions for broader application.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Create Projects

Dear Sir,

With reference to the subject cited above we wish to purchase components for our projects under Create SIG, the digital arts and design SIG of ISTE. For the current semester, we have curated a lineup of interactive digital projects, such as The Final Semester, a 2D game that gamifies the NITK campus experience , and ArtIn, a digital archive for Indian art and textiles built on a no-code framework. We are also designing a 3D Escape Room, a puzzle-based environment that challenges players' problem-solving skills through mystery and exploration. These initiatives are intended to cultivate proficiency in game development, UI/UX design, and digital narrative construction among the students.

Permission is sought to proceed with the development and implementation of these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: THE FINAL SEMESTER - A 2D SURVIVAL GAME

TEAM: 5 MENTORS + 6–7 MENTEES

1. Situation / Problem

Students often find campus-based learning monotonous. There is a need for an interactive, gamebased environment to explore the NITK campus and learn through engagement. The project aims to convert the NITK map into a playable survival game world.

2. Task / Objective

- Build a 2D pixel-art survival game.
- Implement combat, exploration, health, stamina, and inventory systems.
- Create AI enemies with patrol/chase/attack behaviour.
- Use Unity + C# for real-time gameplay.

3. Action / Implementation

Designed the world using Unity Tilemaps based on NITK locations (hostels, academic blocks, library, main road).

Created player and enemy sprites, animations, and behaviour scripts.

Added major mechanics:

• Player movement & attack
• Health & stamina system
• Enemy AI
• Inventory system + checkpoints

Developed UI screens: Start Menu, Pause Menu, and Game-Over interface.

4. Result / Outcome

Built a working survival game prototype with smooth movement, combat mechanics, and AIdriven enemies.

Produced a virtual NITK campus in 2D pixel-art style.

The game is fully play-testable and ready for deployment as a PC/Web demo.

5. Future Scope Multiplayer mode

- Full NITK map (auditorium, beach, canteens)
- Advanced pathfinding (A*)
- Android/iOS mobile version
-

6. Total Project Budget: 0 (Software used: Unity – Free Version, Assets created in house**)**

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE : ARTIN – A NO-CODE INTERACTIVE PLATFORM FOR INDIAN ART & TEXTILES

TEAM: 4 MENTORS + 4 MENTEES

1. Situation / Problem

Indian art knowledge is scattered offline, making it difficult to explore.
ArtIn aims to create a single, accessible digital platform showcasing Indian art while teaching juniors UI/UX and no-code development.

2. Task / Objectives

- Digitally archive Indian art forms.
- Design an interactive website with quizzes and visuals.
- Mentor juniors in UI/UX and no-code tools.
- Build Version 1 in 3–5 months.

3. Action / Implementation Plan

- Research art categories and curate content.
- Create wireframes, user flows, and a design system.
- Build high-fidelity designs and add interactions.
- Convert designs to a no-code website using Framer.
- Test usability and optimize for mobile.

4. Expected Result / Outcome

- A working beta with 3–4 art categories.
- A responsive, interactive no-code website.
- Improved UI/UX and prototyping skills for mentees.

5. Future Scope

- Add more art categories and regional content.
- Multilingual support and museum collaborations.
- E-commerce features for artisans.
- Mobile app version.

6. Total Project Budget : ₹0 — Built using no-code tools and in-house assets (Framer, Figma).

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: ESCAPE ROOM – A 3D PUZZLE GAME

TEAM: 5 MENTORS + 4 MENTEES

1. Situation / Problem

We plan to build a 3D mystery escape-room game in Unity where the player finds hidden objects and solves puzzles to escape. The project helps us learn Unity, C#, and 3D game design.

2. Task / Objectives

- Create a 3D escape-room game.
- Add player movement and object interaction.
- Design rooms, lighting, and puzzles.
- Include animations, sounds, and UI.

3. Action / Implementation

- Build rooms and environment.
- Add player controller + camera movement.
- Implement item picking and door unlocking.
- Add animations and puzzle logic.
- Test and polish gameplay.

4. Expected Outcome

- A working 3D escape-room prototype.
- Functional puzzles, clues, and room transitions.
- Better skills in Unity, C#, and level design.

5. Future Scope

- Expand the platform by adding more art categories and regional textiles.
- Add multilingual support for wider access.
- Enable e-commerce features for artisans.
- Develop a mobile app version (iOS/Android).

6. Project Budget : ₹0 - Unity Free Version + in-house assets.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Credit Projects

Dear Sir,

With reference to the subject cited above, we seek permission to proceed with the projects under Credit, the finance and analytics SIG of ISTE. For the current year, we have curated a lineup of projects that merge financial theory with advanced computing. These include an AI-powered fintech platform for personalized investment insights and a Fundamental Analysis Dashboard that streamlines data for Nifty 500 companies. We are also developing a Quantitative Disruption Index to measure industry risks and a Futuristic Stock Market Index that tracks deep-tech sectors, alongside a Dynamic DCF Model that utilizes Monte Carlo simulations. These initiatives aim to cultivate proficiency in financial modeling, data analytics, and economic forecasting among students.

Permission is sought to proceed with the development and implementation of these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: INDUSTRY DISRUPTION ANALYSIS

Mentors: Mithul, Chethana,Hriday

Mentees: Atharva Banni, Akhand Bhardwaj, P Deekshith, Bharath Chowdary KB, Achutha P

1. Problem

Traditional methods for evaluating industry disruption rely heavily on subjective expert opinion, isolated qualitative insights, and outdated frameworks. With rapid technological evolution (AI, automation, digital infrastructure, deep tech), industries require a **data-driven, quantifiable**, and **continuously updating** model to measure their vulnerability to disruption.The absence of such a model limits strategic planning, innovation forecasting, and competitive analysis.

2. Objective

- Develop a **Quantitative Disruption Index (QDI)** that scores industries using multi-vector datasets.
- Replace subjective, narrative-based disruption analysis with a **rigorous, automated data-focused framework**.
- Build predictive models using adoption S-curves and diffusion models to forecast technological shifts.
- Deploy an interactive **Executive Dashboard** that visualizes disruption levels, trends, and competitive mapping.

3. Methodology.

- Define QDI metrics: Capital Velocity, Technology Penetration, Market Adaptability, Innovation Density, and Public Discourse Signals.
- Develop predictive models for technology adoption using Bass diffusion and trend forecasting.
- Build a 2x2 “Disruption Chart” comparing **Current Disruption Level vs. Future Susceptibility**.
- Construct an interactive Tableau/Power BI Executive Dashboard with drill-downs, industry

4. Expected Outcome

- A validated **Quantitative Disruption Index** applicable across industries.
- A complete, data-backed **market intelligence report** with competitive mapping.
- Predictive insights into technological adoption curves and disruption timelines.
- Identification of high-risk industries and high-growth disruptors.

5. Future Scope

- Expand the QDI model to multiple industries (transport, healthcare, manufacturing, BFSI, etc.) for a macro global disruption map.
- Develop a **real-time Disruption Monitoring Platform** that auto-ingests data and refreshes QDI scores.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: BUILD YOUR OWN FUTURISTIC STOCK MARKET INDEX

Mentors: Hriday jain,Mithul,Aditya jyothir,Aksheet

Mentees: Tejasva Pandey, Vaibhav Porwal, Jay, Adi Nahata, Aditi

1. Problem

Traditional Indian stock indices such as **NIFTY 50** and **Sensex** represent the current economic landscape but fail to capture the **future engines of growth** emerging from AI-driven industries, deep-tech sectors, and rapidly evolving digital ecosystems.

2. Objective

- Create a stock market index representing futuristic, innovation-driven industries.
- Include companies from sectors influenced by AI, automation, sustainability, and deep tech.
- Develop a transparent methodology for stock selection and weighting.
- Backtest the index against NIFTY 50 and Sensex.
- Build a functional dashboard for real-time index tracking.

3. Methodology

- Identify upcoming high-growth industries and shortlist representative companies.
- Design weighting rules (market-cap, equal-weight, or innovation-weight).
- Use APIs to fetch live stock prices and compute index values.
- Build a front-end dashboard showing index movement, constituents, and comparisons.
- Perform backtesting and trend analysis to validate performance.

4. Expected Outcome

- A futuristic index reflecting India's long-term economic growth sectors.
- Accurate comparison between traditional and innovation-driven indices.
- A functional dashboard providing real-time insights.
- Clear documentation supporting stock selection and index logic.

5. Future Scope

- Expand into thematic indices (AI Index, Clean Energy Index, Robotics Index).
- Introduce machine learning for predictive rebalancing.
- Deploy as a public web tool for investors and students.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

PROJECT TITLE: AI-POWERED FINTECH INVESTMENT INTELLIGENCE PLATFORM

Mentors: Shreya Nyamati, Lalima Singh, Abhimanyu Binu, Vinay Kotwani

Mentees: Pranav Shaji, Achutha P

1. Problem

Investors struggle to make data-driven decisions due to scattered financial information, inconsistent insights, and the influence of market sentiment. Traditional tools fail to combine fundamentals, technicals, and behavioral signals into a single intelligent system. Students must learn financial modeling, ML, NLP, and dashboarding to build a unified platform that offers real-time, personalized investment intelligence.

2. Objective

- Build an AI system that provides investment recommendations using ML/DL models.
- Combine financial data, sentiment signals, and portfolio analytics in one interface.
- Implement stock ranking, risk profiling, and trend forecasting.
- Visualize insights using interactive dashboards.

3. Methodology

- Collect historical and real-time data from financial APIs.
- Clean, preprocess, and structure datasets for modeling.
- Develop forecasting models and risk assessment tools.
- Use NLP to analyze news, reports, and social media sentiment.
- Build dashboards to display portfolio health, alerts, and projections..

4. Expected Outcome

- A functional system capable of generating intelligent investment insights.
- Accurate trend forecasting and risk evaluation.
- A dashboard that helps users track portfolios and identify opportunities.
- Personalized recommendations combining data and sentiment.

5. Future Scope

- Add cryptocurrency and alternative asset support.
- Real-time alerting for sentiment, price dips, and volatility shifts.
- Automated AI-based portfolio rebalancing.

6. Total Project Budget: NIL

PROJECT TITLE: DYNAMIC DCF MODELING USING OU PROCESS & MONTE CARLO SIMULATIONS

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Mentors: Bhavya Saini, Hridya Jain, Anantha Krishnan, Chethana Ramesh, Ashmit Singhal

Mentees: Atharva Banni, Harsh Vardhan, Shashank , Shamik Neendoor, Adi Nahata

1. Problem

Traditional DCF valuation assumes a fixed discount rate, which fails in volatile interest rate environments. Real markets experience mean reversion, shocks, and uncertainty that static models cannot capture. Students must learn stochastic finance, Monte Carlo simulation, and quantitative modeling to build a valuation system that reflects real-world rate behaviour and risk dynamics.

2. Objective

- Implement a DCF model using **Ornstein–Uhlenbeck (OU)** mean-reverting interest rate modelling.
- Estimate OU parameters using historical rate data.
- Run Monte Carlo simulations to generate thousands of rate paths.
- Integrate risk analytics such as confidence intervals, VaR, and CVaR.

3. Methodology

- Collect and preprocess historical SOFR/Treasury rate data.
- Implement the OU stochastic differential equation and discretization.
- Estimate parameters (θ, μ, σ) via Maximum Likelihood Estimation.
- Build the Monte Carlo engine for rate-path simulation.
- Integrate simulated rates into a dynamic DCF calculator.
- Visualize outputs using plots and dashboards; benchmark against static DCF.

4. Expected Outcome

- A dynamic DCF model reflecting realistic interest rate behaviour.
- NPV distributions showing uncertainty and valuation ranges.
- Risk metrics such as VaR, CVaR, and confidence intervals.
- Better valuation accuracy for long-term and rate-sensitive assets.
- Validation against traditional DCF outputs.

5. Future Scope

- Add multi-factor stochastic models like Vasicek, CIR, and Hull–White.
- Use machine learning for parameter estimation and regime detection.
- Extend modelling to credit spreads, volatility, and FX rates.
- Build portfolio-level risk and valuation tools.

6. Total Project Budget: NIL

PROJECT TITLE: Fundamental Analysis & Long-Term Investing Dashboard

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Mentors: Prashanth, Karthik, M S S Vinay, Varun, Manas

Mentees: P Deekshith, Bharath Chowdary KB, Joseph V Francis, Diya Zacharia, Rishi

1. Problem

Investors struggle to analyze the fundamentals of Nifty 500 companies due to scattered data, inconsistent formats, and manual comparison across multiple platforms. Tracking valuation ratios, profitability metrics, and growth indicators becomes time-consuming and error-prone. Students must learn automated data collection, Power BI modeling, and financial analysis to build a unified platform for fundamental research.

2. Objective

- Create a centralized dashboard covering fundamental data for Nifty 500 companies.
- Automate collection of 50+ key metrics using Python and APIs.
- Build structured datasets for fast analysis and comparisons.
- Provide insights to identify undervalued and high-quality businesses.

3. Methodology

- Automate data fetching using *yfinance* in batches to extract financial statements, ratios, and historical metrics.
- Clean and transform raw data into standardized CSV formats.
- Import datasets into Power BI and build data models.
- Create interactive dashboards with filters, drill-downs, and sector views.

4. Expected Outcome

- A fully functional Power BI dashboard for analyzing Nifty 500 fundamentals.
- Faster comparison of companies and identification of strong long-term picks.
- Structured datasets usable for future financial research.
- Clear insights into profitability, valuations, growth, and sector strength.

5. Future Scope

- Add automated quarterly updates to keep the dataset fresh.
- Include advanced valuation models (DCF, PE regression, Piotroski score).
- Build a portfolio tracker integrated with real-time price feeds.
- Create an AI-based recommendation layer for long-term investing.

6. Total Project Budget: NIL

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Subject: Permission sought to purchase components for ISTE Crypt Projects

Dear Sir/Madam,

With reference to the subject cited above, we seek approval to conduct the projects under Crypt, the Computer Science SIG of ISTE. We are enthusiastic about diverse computational domains, ranging from artificial intelligence and automation to fintech and data privacy. This semester, we are undertaking projects driven by advanced software engineering and research, including CloudFunc AI and an API Documentation Agent for developer automation , FinSight for financial tracking , and privacy-focused systems involving Ethical AI and Federated Learning. These initiatives aim to foster technical expertise by building innovative solutions and validating algorithms through rigorous testing.

Permission is sought to proceed with the development and implementation of these projects.

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Crypt Project: Cloud Func

Mentors	**Mentees**
Nikhil Agarwal	Kowndinya , Jiya
Lavanya Rathi	Pratham , Nikitha
Atharva Parkhi	Bhavya , Mili
Aaryan Yadav	Santosh , Prashasti

About the Project

CloudFunc AI is an intelligent system that converts **plain-English cloud requirements into deployable serverless function code**. Instead of developers manually searching through AWS/Azure/GCP documentation and configuring triggers, permissions, and services, CloudFunc automates the entire process.

The system crawls cloud documentation, stores it in a **RAG (Retrieval-Augmented Generation) pipeline**, and uses an LLM to understand user requests. It then generates **structured JSON instructions**, which are converted into **ready-to-deploy serverless code** (Python/Node.js/Go) using Jinja2 templates. A secure **sandbox** allows developers to test the generated functions instantly.

CloudFunc AI removes the need to read long documentation pages — developers can simply describe the task in natural language, and the system returns complete serverless functions with correct triggers, inputs, IAM permissions, and environment configurations.

COMPONENTS/BUDGET: NONE

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Crypt Project: LLM

Mentors
Radhika Sharda
Sanstuti Mishra
Pramod Budhiraja

Mentees
Nabeel , Rahul
Umar , Rushi
Pranav

About the Project
The project addresses the problem of messy, unstructured API documentation by building an autonomous agent that ingests natural language docs (HTML/Markdown) and converts them into executable code. Technical Workflow: 1. Ingestion: We use Scrapy to crawl and extract text from documentation websites. 2. RAG Pipeline: The text is chunked using LangChain Splitters and embedded using Sentence-Transformers into a ChromaDB vector store for semantic retrieval. 3. Reasoning: When a user queries, the system retrieves relevant context and uses an LLM to generate a structured JSON object containing the API method, URL, headers, and body . 4. Code Generation: A Jinja2 templating engine converts this JSON into ready-to-run Python or cURL snippets. 5. Interface: The user interacts via a React + TypeScript chat interface, backed by a FastAPI service.

COMPONENTS/BUDGET: NONE

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Crypt Project: FinSight

Mentors	**Mentees**
Sai Chaitanya	Anirudh
Pranav Venkat	Aryan Bokolia
Sai Govardhan	Aryan Sharma
Dodda Karthik	Rohit , Seesa Harika

About the Project

FinSight is like a smart piggy bank that lives on your phone. Lots of grown-ups lose track of their money and feel stressed because they don't know exactly where it goes. Our app fixes this by watching your money for you. It doesn't just write down how much you spent on candy or clothes, it sees your receipts, helps you split the cost of pizza with your friends , and gives you a report card on your spending so you can save more pocket money for your toys. We want to help people stop being "blind" about their money so they can make smart choices.

COMPONENTS/BUDGET: NONE

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

To
Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

17th November 2025

ISTE Crypt Project:Ethical AI

Mentors	**Mentees**
Basitha	Harsha , Pranav
Heemakshi	Varun , Vinayak

About the Project

Machine unlearning in healthcare means training a prediction model on patient data and then being able to make the model forget specific patient records whenever a deletion request is made. After removing the selected data, the model is retrained or adjusted so it behaves as if that information was never used. A dashboard is used to show how the model's predictions and accuracy change before and after the unlearning process. Throughout this setup, all data must be anonymized and handled according to GDPR and HIPAA rules, ensuring patient privacy, secure storage, and ethical use of AI.

COMPONENTS/BUDGET: NONE

To

INDIAN SOCIETY FOR TECHNICAL EDUCATION, NITK CHAPTER

Dean Student Welfare
National Institute of Technology, Karnataka
Surathkal

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ISTE Crypt Project:Federated Learning

Mentors	**Mentees**
Abhimanyu	Dhruva , Emerin
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About the Project

This project creates a privacy-focused NLP system that can identify risky clauses in legal contracts without ever sharing the actual documents. Instead of collecting confidential contracts in one place, it uses Federated Learning, where each organization keeps its data locally and only sends encrypted model updates. This allows a shared model to be trained across multiple clients while fully maintaining privacy and complying with regulations.

The system analyses contracts to find clauses that may be high-risk, vague, biased, or missing important protections. It processes the documents, extracts relevant clauses, converts them into transformer-based embeddings like LegalBERT or FinBERT, and trains the risk-prediction model through a federated setup using frameworks such as Flower or PySyft.

COMPONENTS/BUDGET: NONE